

Deliverable Content for CD.pdf

Pipeline Diagram:

1. **Successful CI run on main branch** (Triggers the CD pipeline) → ↓
2. **deploy-canary** (Pulls Image & Spins up Canary Port) → ↓
3. **canary-analysis** (Health Check Endpoint `?num1=5&num2=7` = 12) → ↓ (If Success)
4. **promotion** (Deploy 100% Traffic, Stop Old Prod Container)

(Note: If Analysis Fails at step 3, the Pipeline Stops and Reports Failure)

Tech Overview: GitHub Actions is used for the orchestration. Deployment is containerized using Docker.

Canary Analysis: The pipeline performs a smoke test by hitting the `/api/health/canary-check` endpoint. It validates that the service returns a specific JSON object confirming `5 + 7 = 12`.

Canary Promotion: After validating the canary image in our analysis, we replace the existing production container with the new image.

Downtime/Impact: Estimated downtime is ~5-10 seconds while the production container restarts with the new image. This impact is minimal as the analysis ensures the new version is functional before the switch.

Link:

- GitHub Actions Workflow: [.github/workflows/cd.yml](https://github.com/workflows/cd.yml)